

WULUNGURRU, Australia

WMO: 943210

Lat: 23.2657S

Lon: 129.3840E

Elev: 1494

StdP: 13.92

Time Zone: 9.50 (AUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	48.1	50.1	5.9	7.9	71.8	9.5	9.4	75.1	23.0	55.7	21.6	57.0	12.2	150	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	108.6	66.3	106.4	66.2	104.3	66.0	76.2	85.7	75.0	85.5	73.9	85.9	10.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	135.0	79.7	72.6	127.6	79.9	70.5	118.8	80.6	40.9	86.1	39.8	85.1	38.7	86.3	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	21.7	20.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	91.2	89.4	87.4	80.2	70.3	63.4	63.5	67.1	75.4	82.9	85.6	89.4
	DBStd	11.53	5.98	5.24	5.80	5.69	5.70	4.70	5.00	5.42	6.23	5.98	6.04	6.09
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	242	0	0	0	0	18	84	89	43	7	1	0	0
	CDD50	10492	1276	1103	1158	905	628	402	420	531	761	1020	1068	1220
	CDD65	5261	811	683	693	455	182	36	44	109	319	556	618	755
	CDH74	72185	12821	10400	9942	5017	1306	203	386	998	3385	7314	8828	11585
CDH80	43532	8666	6722	6069	2462	383	25	51	252	1655	4242	5406	7599	
Wind	WSAvg	11.8	11.8	11.7	12.2	12.2	11.6	11.5	11.2	11.3	11.7	11.9	12.2	11.8
Precipitation	PrecAvg	11.2	1.7	2.0	1.1	0.6	0.7	0.5	0.4	0.2	0.2	0.6	1.1	2.0
	PrecMax	29.4	5.0	13.8	8.1	4.0	3.6	2.9	2.5	1.6	2.8	3.0	3.2	8.1
	PrecMin	1.8	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2
	PrecStd	6.0	1.1	2.6	1.8	0.9	0.9	0.6	0.6	0.3	0.5	0.7	0.8	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	111.8	108.5	106.7	100.3	90.3	83.3	83.7	89.3	97.9	103.7	107.0	111.7
		MCWB	67.7	65.8	65.7	65.6	59.7	58.4	56.5	58.2	60.3	62.4	64.6	66.9
	2%	DB	108.9	106.0	104.1	97.1	87.0	79.0	81.3	85.8	95.4	101.7	104.1	108.4
		MCWB	66.9	66.1	66.0	62.8	59.9	57.8	56.0	57.1	59.5	61.8	64.1	66.1
	5%	DB	106.3	103.8	101.7	94.8	84.3	75.9	78.8	82.7	92.6	99.8	102.0	105.8
		MCWB	67.0	66.1	65.9	62.5	59.3	56.0	55.0	55.0	58.6	61.4	63.8	66.0
	10%	DB	103.9	101.4	99.2	91.7	81.5	72.9	75.2	80.0	89.4	97.3	99.2	102.6
		MCWB	67.3	66.1	65.2	61.5	58.0	54.0	53.6	54.7	57.7	60.9	63.3	66.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	76.5	76.5	71.0	69.2	65.9	64.1	64.6	66.5	69.9	71.1	76.5
		MCDB	85.0	84.7	84.7	82.2	71.8	72.9	74.8	72.4	75.1	82.7	88.8	84.1
	2%	WB	76.6	75.4	74.6	68.9	65.9	62.2	61.0	60.7	63.8	67.7	70.2	74.5
		MCDB	86.7	86.2	84.6	81.8	74.6	71.9	73.8	77.0	80.5	81.9	88.6	84.4
	5%	WB	75.4	74.5	72.8	66.9	63.1	59.2	58.4	58.2	61.1	66.4	69.3	73.4
		MCDB	85.9	86.7	86.4	84.8	77.9	69.7	69.8	78.3	84.4	81.5	88.1	85.9
	10%	WB	74.1	73.2	71.3	64.7	61.0	56.8	55.6	55.8	59.2	64.8	68.2	72.1
		MCDB	85.6	87.1	87.0	83.9	76.4	69.7	70.6	77.4	85.8	82.1	87.2	87.6
Mean Daily Temperature Range	5% DB	MDBR	20.7	20.5	19.0	19.2	19.0	18.1	20.7	22.5	23.9	23.6	22.9	21.5
		MCDBR	22.9	22.2	21.8	21.4	21.3	20.9	24.5	26.3	26.9	26.9	25.0	24.0
		MCWBR	6.8	6.4	6.9	7.3	8.4	9.0	10.5	11.4	10.0	9.8	8.0	7.6
	5% WB	MCDBR	18.6	19.0	17.2	19.4	17.3	17.4	19.9	22.6	23.0	21.0	22.8	21.0
		MCWBR	5.1	5.4	5.4	6.5	7.9	8.4	9.2	10.9	9.5	7.2	6.2	5.8
Clear-Sky Solar Irradiance	taub		0.360	0.359	0.347	0.328	0.310	0.294	0.300	0.310	0.350	0.357	0.366	0.365
	taud		2.523	2.526	2.545	2.538	2.533	2.560	2.518	2.453	2.371	2.389	2.428	2.489
	Ebn at Noon		312	308	303	296	289	288	289	297	297	305	308	311
	Edh at Noon		36	35	33	30	28	26	28	33	38	40	39	37
All-Sky Solar Radiation	RadAvg		2416	2282	2031	1798	1479	1335	1429	1765	2081	2316	2407	2327
	RadStd		147	170	142	106	125	107	104	73	104	131	116	171

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page